

# torch.broadcast\_shapes#

[https://pytorch.org/docs/stable/generated/torch.broadcast\\_shapes.html](https://pytorch.org/docs/stable/generated/torch.broadcast_shapes.html)

## Description:

Similar to `broadcast_tensors()` but for shapes. This is equivalent to `torch.broadcast_tensors(*map(torch.empty, shapes))[0].shape` but avoids the need create to intermediate tensors. This is useful for broadcasting tensors of common batch shape but different rightmost shape, e.g. to broadcast mean vectors with covariance matrices. `*shapes` (`torch.Size`) – Shapes of tensors. A shape compatible with all input shapes.

## Function Signature

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```
torch.broadcast_shapes(*shapes) → Size[source]#
```

Parameters

Returns

Return type

Raises

**Returns:**

shape (torch.Size)

## Code Examples

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```
>>> torch.broadcast_shapes((2,), (3, 1), (1, 1, 1))  
torch.Size([1, 3, 2])
```

# Detailed Documentation

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A shape compatible with all input shapes.

`RuntimeError` – If shapes are incompatible.

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